



**Technological Institute of the Philippines
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PDF Sense: Digital Reader System

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Introduction

PDF Sense

- **Store** thesis in a centralized repository.
- **Checker** thesis to deduce if the uploaded pdf file is actually a computer science thesis.
- **Analyze** academic papers to create a title and summary for uploaded thesis.

Background of the Study

Research plays a pivotal role in the process of developing a thesis, as it serves as the foundation for supporting arguments, and validating findings of the thesis. One of the most essential parts of research writing is the review of related literature and studies, which requires students to examine a great number of academic studies, journals, and other previously published theses. In this process, researchers gain insights into existing works to identify research gaps, and establish the potential significance of their study. However, looking through a number of academic materials is often a tedious and time-consuming task.

Students frequently encounter difficulties when reading lengthy research documents, particularly thesis papers in PDF format that may contain hundreds of pages as well as confusing terms. Extracting the main ideas, identifying significant findings, and understanding complexities require significant effort and time. As a result, students struggle to efficiently comprehend multiple studies within limited academic deadlines which could potentially result in lower quality of work. Manually summarizing research papers increases the risk of overlooking essential information.

Another challenge faced by students is the organization and compilation of collected studies for the thesis. Managing multiple files, locating relevant sections, and synthesizing information from various sources can become overwhelming, especially when handling numerous documents

simultaneously which could potentially result in some crucial information being forgotten. These challenges highlight the need for a tool that can assist researchers in processing academic materials more efficiently.

With the rise of artificial intelligence, automated text analysis and summarization systems have emerged as a solution to information overload. AI-powered summarization tools can analyze documents, and generate concise summaries while preserving the main ideas of the original study. Such technology has the potential to reduce the workload of students and improve efficiency during the research process.

In response to these challenges, this study proposes the development of CloudPDF: a Web-Based Thesis PDF Summarizer using NodeJS, a website designed to automatically generate meaningful summaries of an uploaded PDF of a thesis. The system aims to help students quickly understand the core concepts of the research paper, resulting in the enhancement of research efficiency. By integrating automated summarization into the research workflow, the proposed system aims to simplify document analysis, improve organization of research materials, and help students in completing their academic requirements more effectively..

Statement of the Problem:

Reviewing related literature is a crucial part of conducting research. However, students often face difficulties when dealing with lengthy research papers in PDF format. The manual process of reading, analyzing, and summarizing these documents is time-consuming, tedious, and may lead to missed information, especially when multiple sources are involved in the process. Therefore, there is a need for a system that can automatically analyze thesis PDFs and generate concise summaries to help students efficiently understand and manage academic research materials.

Specific Problem:

- Students spend significant time manually reviewing and summarizing research papers by hand.
- Students lack a centralized platform to store and manage thesis documents efficiently.
- Students have difficulty quickly identifying relevant information from multiple research sources.

General Objective:

The study aims to develop a web-based thesis pdf summarizer using NodeJS. It can automatically analyze academic documents in PDF format and generate summaries. The system is designed to assist students in efficiently reviewing research materials, resulting in reducing the time and effort required to analyze lengthy thesis papers

Specific Objective:

- Develop an AI assisted summarizer to lessen the workload of the user in summarizing text
- Store thesis and research documents in individual accounts.
- Analyze academic papers to generate a summary for uploaded thesis.

Technology Used:

- Android Studio
- Java
- Firebase

Progress Updates**1. Homepage Design**

- Navigation bar implemented (Home, About, Login, Register).
- Branding “AppDev” added with gradient background.
- The main section includes project title “TELECOMMS,” tagline, and “Get Started” button.
- Footer structured with company info, quick links, and contact details.

Front-End initial design

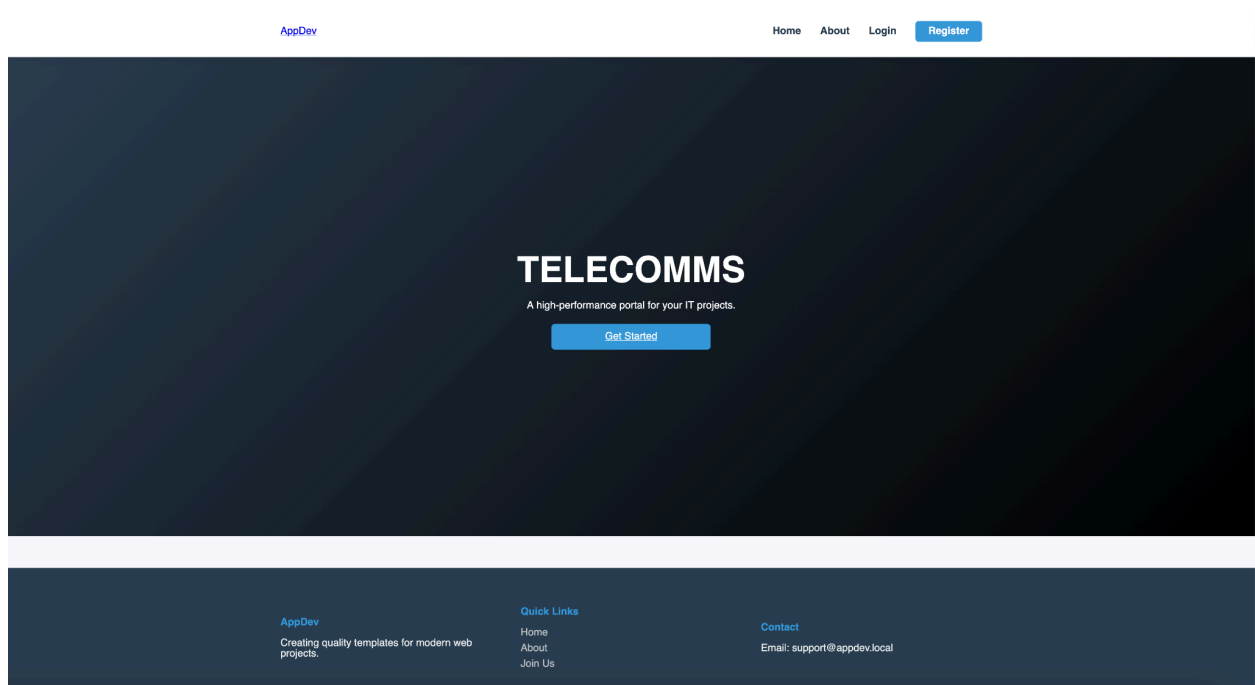


Figure 1. Homepage

. Login Page Design

- Login interface created with branding and navigation bar.
- Centered form includes username and password fields with “Login” button.

- Footer finalized with company info, quick links, contact details, and copyright note.

The screenshot displays the login interface for AppDev. At the top, a light blue header contains the AppDev logo on the left and navigation links (Home, About, Login, Register) on the right. The main content area is a light blue gradient. In the center, a white login form is displayed with a 'Welcome Back' heading. Below the heading are two input fields labeled 'Username' and 'Password', and a blue 'Login' button. The footer is a dark blue gradient. It contains three columns of text: the AppDev logo and tagline 'Creating quality templates for modern web projects.' on the left; 'Quick Links' with links to Home, About, and Join Us in the center; and 'Contact' with the email 'support@appdev.local' on the right. A copyright notice '© 2026 AppDev. All rights reserved.' is centered at the bottom of the footer.

Figure 2. Login

Register Page

- Registration form developed with input fields for username, email, and password.

- “Create Account” section designed with a blue “Register” button.
- Footer includes company info, quick links, and contact details.

The screenshot displays a web application interface for user registration. At the top, a navigation bar includes the 'AppDev' logo, links for 'Home', 'About', and 'Login', and a prominent blue 'Register' button. The main content area is a light purple gradient, centered with a white 'Create Account' form. This form contains three input fields labeled 'Username', 'Email', and 'Password', followed by a blue 'Register' button. The footer is a dark blue section divided into three columns: the first column shows the 'AppDev' logo and tagline; the second column lists 'Quick Links' (Home, About, Join Us); and the third column provides 'Contact' information (Email: support@appdev.local). A copyright notice '© 2020 AppDev. All rights reserved.' is centered at the bottom of the footer.

Figure 3 Register

Login Session

- **Successful login session implemented with personalized greeting (“Hello, caleb26!”).**
- **Navigation bar updated with “Logout” option.**
- **The main panel displays the project title “TELECOMMS” and tagline.**
- **Footer remains consistent with other pages.**

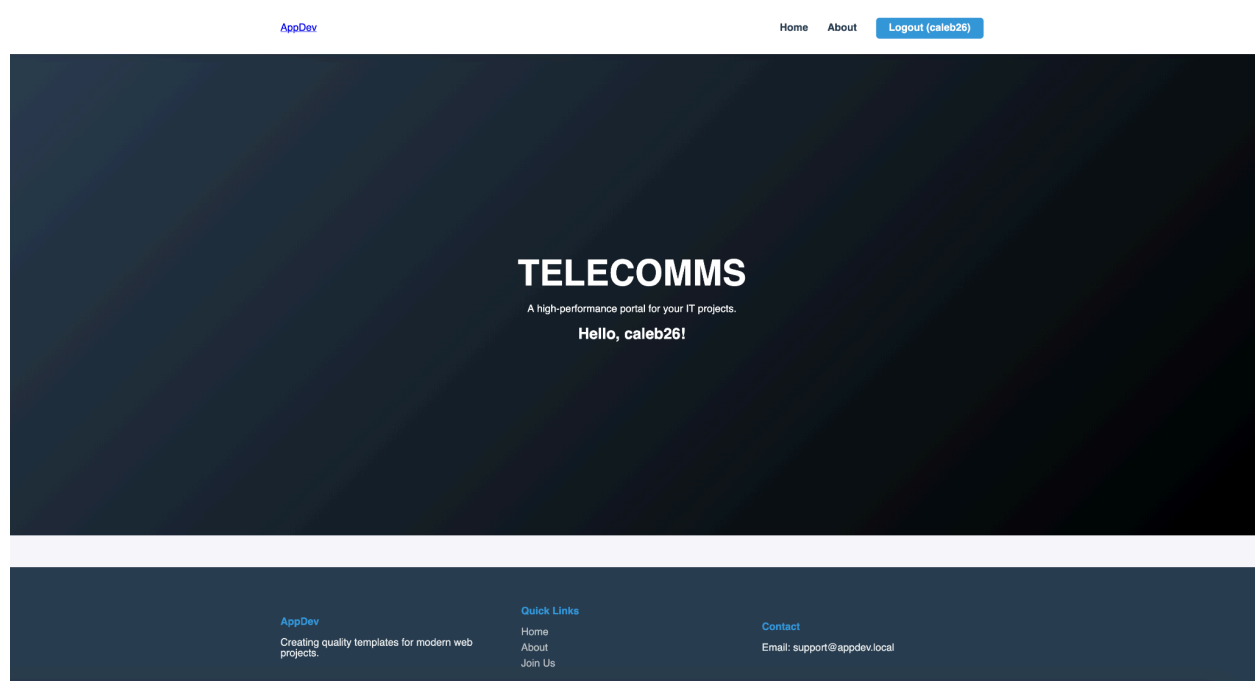


Figure 4 Login Session.

AI Summarizer:

PDF Uploaded Successfully!

File: Tablet Group 3.pdf

Summary (AI Generated)

A tablet is defined as a small, portable, thin device with a touchscreen interface, larger than a typical smartphone but smaller than a laptop. Its key features include a touchscreen, portability, connectivity, multimedia and productivity capabilities, and various operating systems.

The history of tablets dates back to 1972 with Alan Kay's conceptual "Dynabook," which envisioned a thin body, stylus, and high-resolution display. The first tablet on the market was the GRiDPad in 1989, enabling creative user interfaces. Personal Digital Assistants (PDAs) gained traction in 1991, offering sufficient processing for many applications. The "first true tablet" came in 1994 with the Fujitsu Stylistic 500, featuring a pen and Windows 95. It featured an Intel processor, designed to bring desktop-like functionality to a mobile slate form factor.

Comparing a 1984 Fujitsu Stylistic 500 to a modern (2023) Apple iPad (11th generation) highlights massive technological advancements:

- * **Processor:** From a basic Intel 286 to an advanced Apple M1/M2 with multi-core CPU, GPU, and Neural Engine.
- * **RAM/Storage:** From 1 MB of RAM and 10 MB of storage to 16 GB RAM and 1 TB of storage.
- * **Camera:** Absent in early models, but modern tablets feature high-resolution front and rear cameras.

Advantages of tablets include a larger screen for better media consumption and reading, longer battery life compared to smartphones, an excellent app experience for productivity (e.g., Office, PDF), more comfortable typing, and robust accessory support like styluses and detachable keyboards.

However, tablets also have **disadvantages**: they are less portable than smartphones, less convenient for quick tasks like calls or texts, can be slower than dedicated machines for demanding apps despite similar mobile processors, may be overkill for simple tasks like casual browsing, and often require expensive additional accessories like styluses or keyboards.

PDF Parser:

Extracted Text

TABLET
GROUP 3

WHAT IS A
TABLET?

TABLET
It is a small portable thin device
with a touch screen interface
similar to a smartphone,
however, it is bigger than the
typical smartphone but not
bigger than laptops.

FEATURES OF A TABLET
Touchscreen Interface
Portability
Connectivity
Multimedia & Productivity
Operating Systems

HISTORY OF
TABLETS

1972
Alan Kay came up with the concept
of a tablet, originally called
dynabook. It featured a stylus pen
and featured a thin body with a
display that boasted at least a
million pixels.

1989
GRiDPad, the first tablet in the
market. It enables the creation of
different user interfaces such as
animation, drawing, and graphics.

1991
Personal Digital Assistants (PDA), it
has sufficient processing speed and
graphics at the time, and it could
sustain a generous amount of
applications which is why it is widely
used by many companies such as
Nokia.

1994
The first true tablet, Stylistic 500-
tablet by Fujitsu. Image of Alan Kay,
it featured a pen run Windows 95. It
featured an Intel processor,
designed to bring desktop-like
functionality to a mobile slate form
factor.

2010
iPad, it offered everything the users
expected in Kay's vision of the
Dynabook. It ran in iOS, which
allowed the integration of easy-to-
customize features, an intuitive
touch screen, and the use of
gestures.
